Gold and BTC work update

One section per point of my update email. I'm not making the calls here — this is the research and my suggested treatments, prepared for the collaborative process. The feedback I'm asking for: the gold model, the Bitcoin approach, and the two one-pagers (attached).

The update email, point by point — and the specific feedback I'm asking for

Structure

Point from the update	Slides
1 · Full literature reviews and a summary of the most important points	4–5
2 · Translation from the reviews into my view of the key drivers, gold and BTC	6
3 · Initial gold return models with robustness checks — simplicity, assumptions from our existing CMAs	7–8
4 · The appreciation not captured by the model, the error term, qualitative overlays	9
5 · Research opportunities to go deeper (where I would especially value your feedback)	10
6 · Gold correlations and performance during drawdowns (especially left-tail)	11–12
7 · Bitcoin and the inconstancy of its relationship with various signals	13–15
Feedback asks & materials	3, 16
Appendix: frequencies · swap-spec check · holders · valuation framework · official-sector flow · BTC-vs-gold consistency	17–23

- The asks are **feedback on suggested treatments**, not sign-offs — the calls belong to the group process.

The six feedback areas

#	My suggested treatment	The open question
1	Drift = structural growth band, 4.8–6.7%/yr by anchor	which anchor to carry into the process
2	Valuation overlay on the ~20y share cycle	functional form and size
3	One floating DM dollar leg; buyer basket monitored	regime-dependent leg worth the complexity?
4	US 10y real as the rate leg	too dollar-centric? global real rate next study?
5	CPI regime threshold at 4%	vs 2.5%; live cell in the workbook
6	BTC: scenario grid + survival mixture	grid as distribution vs one weighted point

Source(s): Update email 2026-07; slide references at right.

What I'm seeing: gold has a stable factor structure plus an unexplained drift; Bitcoin has no consistent model at all

• **Gold:** three variables already in our CMAs explain ~1/3 of monthly variance and beat a random walk out of sample — the open item is the **drift**, which the factors cannot carry.

Gold

0.30

monthly adjusted R^2, M3 (quarterly 0.42); betas stable in rolling 5y windows

~6%/yr

drift implied by the constant-share identity: asset-pool growth – gold supply growth (band 4.8–6.7 by anchor)

t ≈ 0.1

the mean-reversion term, tested: dead in sample, worse out of sample

• **Bitcoin:** I ran twelve candidate drivers through the identical battery gold passes — **nothing exogenous holds**, so I suggest sizing BTC against gold rather than modeling it standalone.

Bitcoin

0 of 5

factors keeping sign & significance across BTC subsamples (gold, same test: 2; its real-rate t runs -3.1 / -5.8 / -6.7 / -2.3)

3–60%

the honest width of the terminal share of gold's pool — the named unknown in the framework

• **Hedging:** the left-tail evidence is where gold earns the allocation — and the strongest case is **bond bears**, not equity crashes.

Left tail

4/4

equity drawdowns >20% since 1970 with gold positive

+3.1 vs +0.6

gold's median 20y real return (%/yr): equities' bottom-quartile windows vs all windows

+95%

gold through the 2020–23 Treasury bear (+64% through Volcker's)

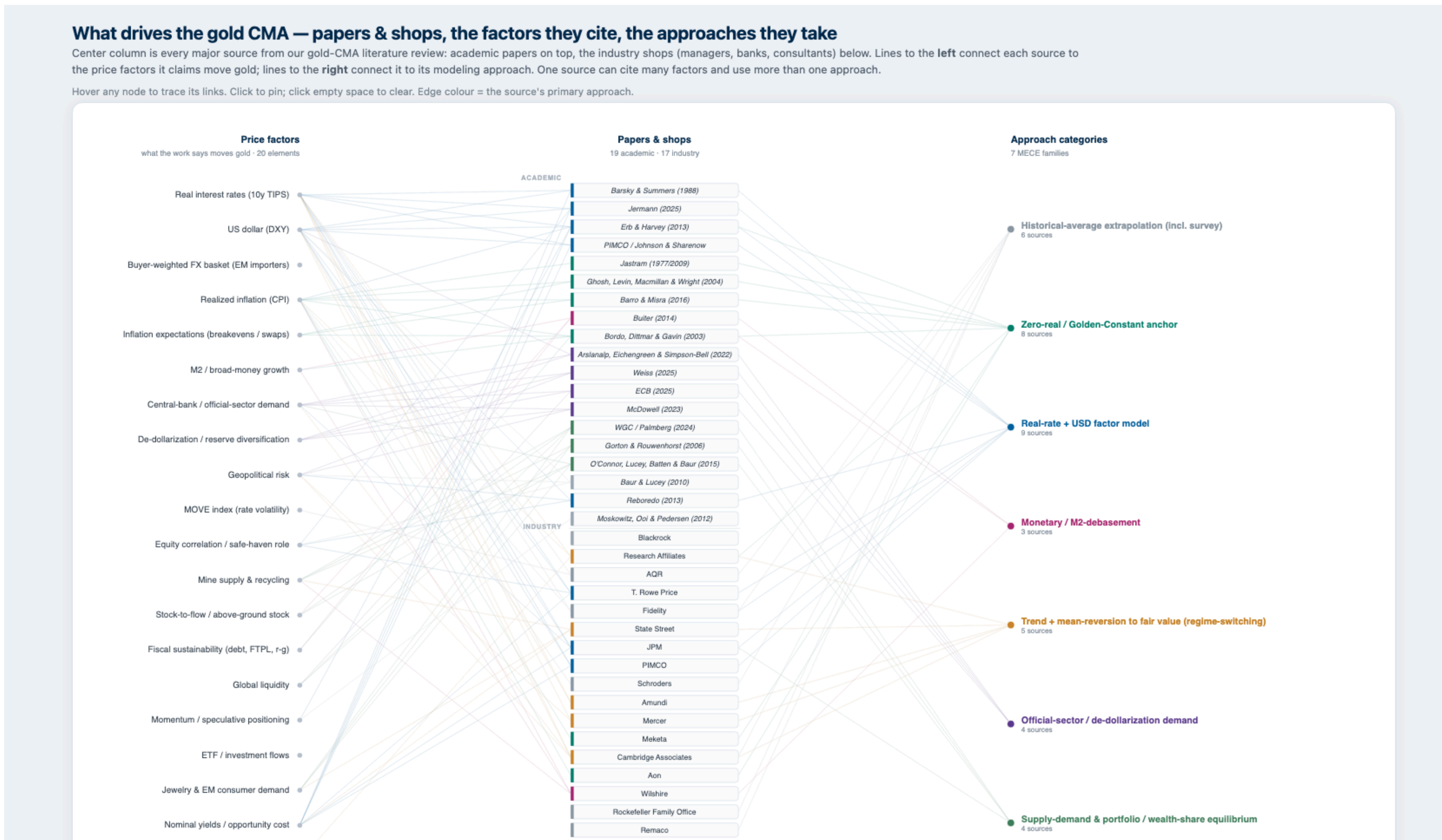
Source(s): Factor workbook (attached); intercept anchor study; portfolio-role battery; BTC driver-instability study.

1 I mapped 19 academic papers and 17 shops to the factors they cite and the approaches they take — 7 families cover everything

You can experiment with the below tool here: jhdavis789.github.io/work-doc-portal/gold-cma-map/cma_map.html

• Center column is every source; left links = the factors it says move gold; right links = its modeling approach. **Real-rate + USD factor models** and **zero-real anchors** dominate; almost nobody models the official-sector buyer.

Papers & shops → price factors → approach families



1 The evidence converges on real rates, the dollar, and long-horizon inflation pass-through; the street ranges ~0% real to ~7% nominal with no consensus

• My read of the academic side: a short list, and an anchor that broke in 2022.

Academic literature — the five results that survive

- **Real rates are the engine** — gold trades like a long-duration real bond; the fit **broke after 2022** (gold doubled while real yields rose).
- **The dollar is the steadiest relationship** — and it held through 2022.
- **Inflation passes through only at long horizons**; the 1–20y hedge evidence is weak.
- **Official-sector demand broke the fixed-mean models** — price-insensitive, sanctions-linked, unforecastable as a factor.
- **Nothing beats a random walk on the level** — VAR, GARCH, ML, stock-to-flow all fail out of sample.

• What the shops do with it: mostly nothing — gold rides in a commodity bucket, and where a number exists the methods disagree.

Published gold CMAs across shops

Shop	Gold CMA
Research Affiliates	~0% real
AQR	1.4% real
State Street	2.5%
JPM	5.5%
Amundi	6%
Mercer	3.4%
Meketa	~0% real
Aon	~0% real
Wilshire	7%
Remaco	1%

Source(s): Lit review (one-pagers attached; full paper 47pp); shop numbers from published CMA documents.

2 My read: three variables carry gold (real rates, dollar, forward inflation); the same exercise for Bitcoin returns empty

• Gold — the set that survived my testing, with the state-dependence made explicit rather than averaged away.

Gold — five factors + one drift

- **Real 10y (TIPS)** — the engine; $\beta \approx -8.3\%$/pp monthly (M3).
- **Dollar** — one floating DM leg; which one is slide 10.
- **Breakevens / inflation swaps** — the forward leg; realized CPI adds nothing.
- **MOVE** — rate-vol insurance bid; second-order but significant.
- **High-CPI regime × real rate** — gold's real-rate beta collapses in uncontrolled inflation; I impose the interaction because least squares underweights rare states.
- **Official-sector drift** — real and structural; belongs in the intercept treatment (slide 9), not the factor list.

• Bitcoin — after the identical translation exercise, nothing exogenous survives; that drives the whole BTC approach.

Bitcoin — the same test, empty result

- **Equity/liquidity beta** is the only supported link — risk appetite, cycle-scale, not a decade drift.
- **Network adoption** is the only defensible drift and it is reflexive — price pulls in users.
- **Inflation hedge, stock-to-flow, on-chain valuation:** not supported in the data.
- Consequence: gold gets a factor CMA; for BTC I suggest a share-of-gold framework with a survival branch (slides 13–15).

Source(s): One-pagers (attached) carry the full translation with citations.

3 A three-variable model from our existing CMA building blocks does most of the work; sophistication adds the regime term; mean reversion adds nothing

• M1 uses only variables the firm already forecasts — the point of building from our existing CMAs. M6 is the mean-reversion test: **gap t = +0.08 monthly**, and walk-forward OOS deteriorates (16% vs 26%).

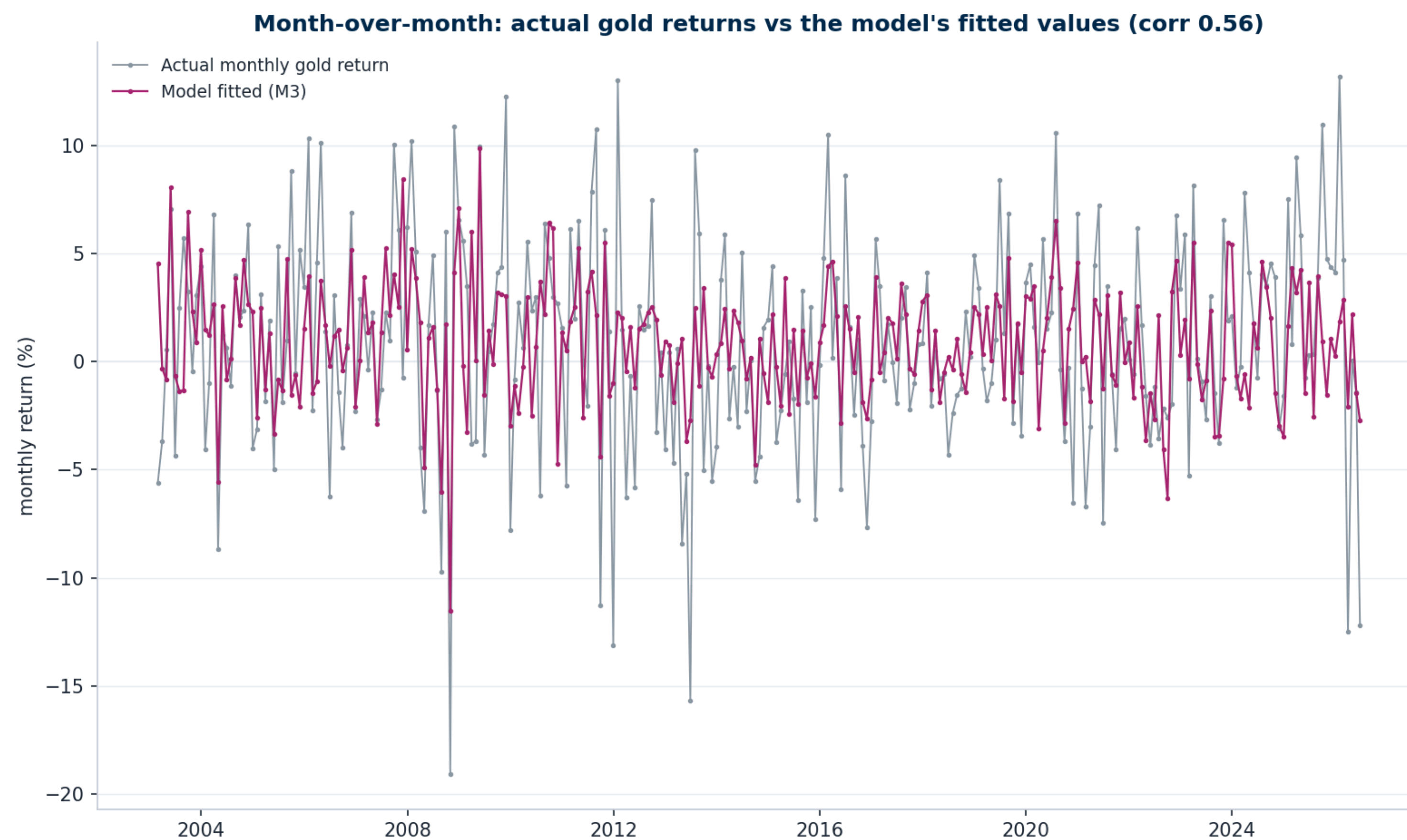
Monthly regression — LBMA PM gold log returns, 2003→

	Dependent variable: gold return (log, %, monthly)		
	M1	**M3**	**M6 (M3 + mean reversion)**
Δ Real 10y (per +1pp)	-6.061*** (-4.97)	-8.292*** (-5.81)	-10.976*** (-6.61)
DXY return (per +1%)	-0.738*** (-5.80)	-0.788*** (-6.33)	-0.754*** (-5.47)
Δ 10Y breakeven (pp)	1.399 (0.87)	2.395 (1.49)	1.889 (1.09)
Δ MOVE (bp)		0.070*** (4.05)	0.088*** (4.53)
Δ Real10 × High-CPI		2.963 (1.23)	4.534* (1.75)
Gold gap vs 5y avg, lagged			0.001 (0.08)
Constant (α, annualized %/yr)	10.27	9.92	9.27
Observations	281	281	222
Adjusted R²	0.260	0.301	0.338

*t-statistics (OLS) in parentheses. *p<0.10, **p<0.05, ***p<0.01. HAC inference in the paper.*

• Month-over-month, the model's fitted values track the actual moves — the factor complex is real, not an averaging artifact. Robustness and out-of-sample: next slide.

Actual vs fitted, monthly (M3)



3 Robustness and out-of-sample: every specification produces positive OOS R^2; the simple structures win or tie; the mean-reversion model is worst everywhere

Rolling 5-year re-fit, predict the next period — OOS R^2 vs the OOS mean

OOS R^2 (vs OOS mean)	M1	M2 (M3 ex-regime)	M4 2Y swap	M5 5Y swap	M6 (M3 + reversion)
Monthly	20.2%	25.6%	22.6%	22.2%	16.3%
Weekly	17.3%	19.3%	18.0%	17.5%	20.2%
Quarterly	37.5%	33.4%	32.7%	35.1%	15.7%

• Given the factor moves, the model beats a random walk at daily / weekly / monthly (Diebold–Mariano $p \leq .016$, full model, not subset-selected); quarterly is a wash (n too small). Rolling 5y betas keep sign throughout (workbook + dashboard); signs hold at all frequencies (appendix A1); adjusted R^2 rises with horizon (0.30 monthly → 0.42 quarterly).

Robustness — what held, what failed, how to read the OOS

- **Held:** real-rate and dollar betas keep sign in every rolling 5-year window; the regime interaction is significant at monthly and quarterly.
- **Failed testing:** mean reversion ($t \approx 0$, OOS worse) · levels/fair-value anchors (permanent sell since 2022) · demand-weighted FX baskets (managed currencies: weight without variance) · rolling the regime term (undefined in low-CPI windows).
- M3 rolls as M2 (a 5y window can hold zero high-CPI months → regime term is full-sample only); M6 rolls ex-regime. Swap specs start 2004-07; M6 needs a 5y warmup — hence smaller OOS samples.
- Lagged factors do NOT forecast gold — the model is a transfer function for forecasted factors, not a timing rule.

Source(s): JAOD analysis — gold_models_bbg.xlsx OOS blocks (validation: 278,749 cells, 0 errors; 256/256 checks vs statsmodels).

4 The error term: what should a store of value keep up with? CPI, money and asset aggregates imply drifts of +2 to +7%/yr

- All the factors mean-revert, so the trend lands in the constant — the model-era intercept is ~10%/yr (2003→, the window TIPS data allows). The candidates for what a store of value should keep up with imply different structural rates; gold's realized 1971→ drift is +8.4%/yr.

Which aggregate does the drift track? (quarterly, 1971→)

Anchor	Growth %/yr	Implied drift*
M2	+6.4	**+4.8**
CPI	+3.8	**+2.2**
Nominal GDP	+6.1	**+4.4**
Total debt (all sectors)	+7.7	**+6.0**
Federal debt	+8.4	**+6.7**
Total financial assets	+7.8	**+6.1**

*anchor growth − 1.66%/yr stock growth; ≈2.3%/yr real. Reversion half-lives run 6–70y (~1–4 independent episodes in 55y): the tether is an assumption, not an estimate — which is why the model lives in returns space.

- Considerations rather than conclusions — this is feedback area #1–2.

Open considerations on the drift

- **CPI** has measurement issues (hedonic adjustment) and prices consumption; a store of value arguably insures *wealth*. One line of thinking: demand for the hedge scales with the pool of things that appreciate — implying drift ≈ asset growth − supply growth (my working numbers: ~6.5% global asset growth − ~1.5% supply ≈ 5%/yr).
- **M1** is unusable after the May-2020 redefinition; **M2** is the only anchor with statistical mean reversion (half-life ~6y). The rest have 27–70y half-lives — under one independent episode in the sample, so any tether is an assumption, not an estimate.
- Two cycles sit on the sample: gold's share of US financial assets ran 32% (1980) → 4% (2000) → 97th percentile today, and the estimation window spans a **political cycle** from post-war calm to two live conflicts plus China risk. How much of the ~10%/yr sample intercept is structural vs cyclical is exactly the open question.
- Possible treatments for the group: a fixed constant · a growth band by anchor (4.8–6.7%/yr) · a band plus a valuation/political overlay.

Source(s): JAOD analysis — intercept anchor study (chartpack, incl. the 150-year view); FRED Z.1 + JST Macrohistory; WGC stock anchors.

5 Question to go deeper on: are DXY and US real rates too Western-centric? Today gold trades mainly with the USD — how that could evolve is what I want to research

• The question, via the owner-utility lens you suggested: a trade-weighted dollar is neither the marginal buyer nor the average holder of gold. The work: nine USD measures raced 1990–2026, including buyer-weighted and holder-weighted baskets. **Conclusion today: gold trades with the floating USD** — the Fed major index encompasses DXY (t −3.7 vs +1.0); demand weights fail mechanically (CNY/INR carry the weight but are managed: weight without return variance).

Which dollar? Monthly gold returns, HAC

Univariate R^2	DXY	Fed major	Buyer (flow)	Holder (stock)	Official holder
1995–2026	0.178	**0.208**	0.132	0.165	0.114
2000s	0.248	**0.310**	0.168	0.226	0.219
2010s	0.099	**0.121**	0.107	0.110	0.085
2020s	0.264	0.242	0.220	**0.269**	0.221
2022+	0.253	0.235	0.284	**0.304**	0.213

Shaded = best in window. The marginal buyer and the average *private* holder turn out to be the same ~60% China+India object; the official-holder basket is Euro-heavy (the US hoard drops out as numeraire) and adds nothing.

• The research question: **how could this change over time?** The buyer and holder base is shifting East, and since 2022 the Asia basket leads for the first time. Same question for rates: are US real rates the right opportunity cost, or a Western-centric choice?

Today vs how it could evolve

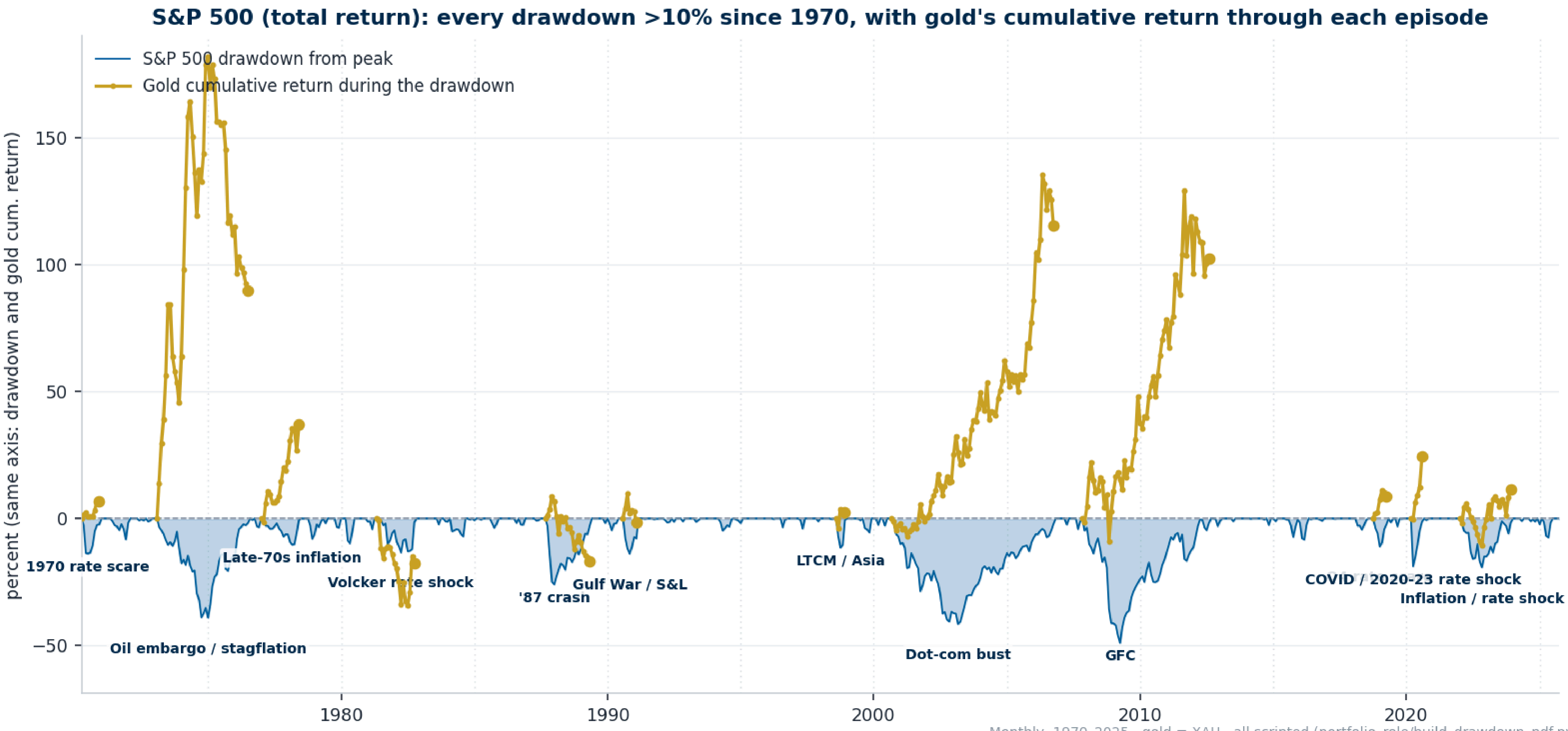
- **Today:** not a trend yet. The Asia holder/buyer basket wins only in the 2022+ window (R^2 .304, the one significant addition); a 10-year rolling test reads it as a short-window regime feature (t −1.9), and an identical signal around 2000 decayed to zero. I carry one floating DM leg and re-test the basket yearly.
- **Rates:** interest-rate parity would say one rate suffices — but the marginal buyers' currencies are managed, so parity fails exactly where the demand is. A GDP-weighted global real rate in the same encompassing design is the natural next study; untested so far.
- **Why it matters:** if the 2022+ signal holds on five years of data, a regime-dependent dollar leg becomes defensible — and it changes the beta applied to the firm's dollar path.

Source(s): JAOD analysis — USD-measure study + holder-basket extension (one-pager); FRED H.10, OECD MEI, WGC/IMF holdings.

6 Gold finished up in 4 of 4 equity drawdowns deeper than 20% since 1970 — and the failure mode (rate shocks) is specific and priced by the regime term

• Protection improves with severity. The strongest and least-appreciated result is **bonds**: gold +95% through the 2020–23 Treasury bear, +64% through Volcker's. The one failure is 2022-style rate shocks — surging real rates cancel the gold bid, which is exactly what the regime interaction prices.

Every S&P 500 drawdown >10% since 1970: index drawdown (blue) and gold's cumulative return through the episode (gold line), same axis, episodes labeled



Source(s): JAOD analysis — portfolio-role battery, monthly 1970–2025. All four indices (S&P, small-cap, 60/40, Treasuries): chartpack.

6 In the worst quartile of S&P 5-year windows (nominal, 1976–2025), gold's median 5-year return is +18%/yr vs +3.7%/yr across all windows

You can experiment with the below tool here: jhdavis789.github.io/gold-holders-dashboard/

• How to read this: each row is one asset; every faint dot is one overlapping 5-year window's annualized nominal total return (box = interquartile range, bar = median). The **bold dots mark the same calendar windows in every row** — the ones where the S&P delivered its worst-quartile 5-year outcome — so each row shows what that asset did while equities had their worst runs. **Red boxes: focus on equities vs gold** — gold's bold dots sit far to the right of its own distribution.

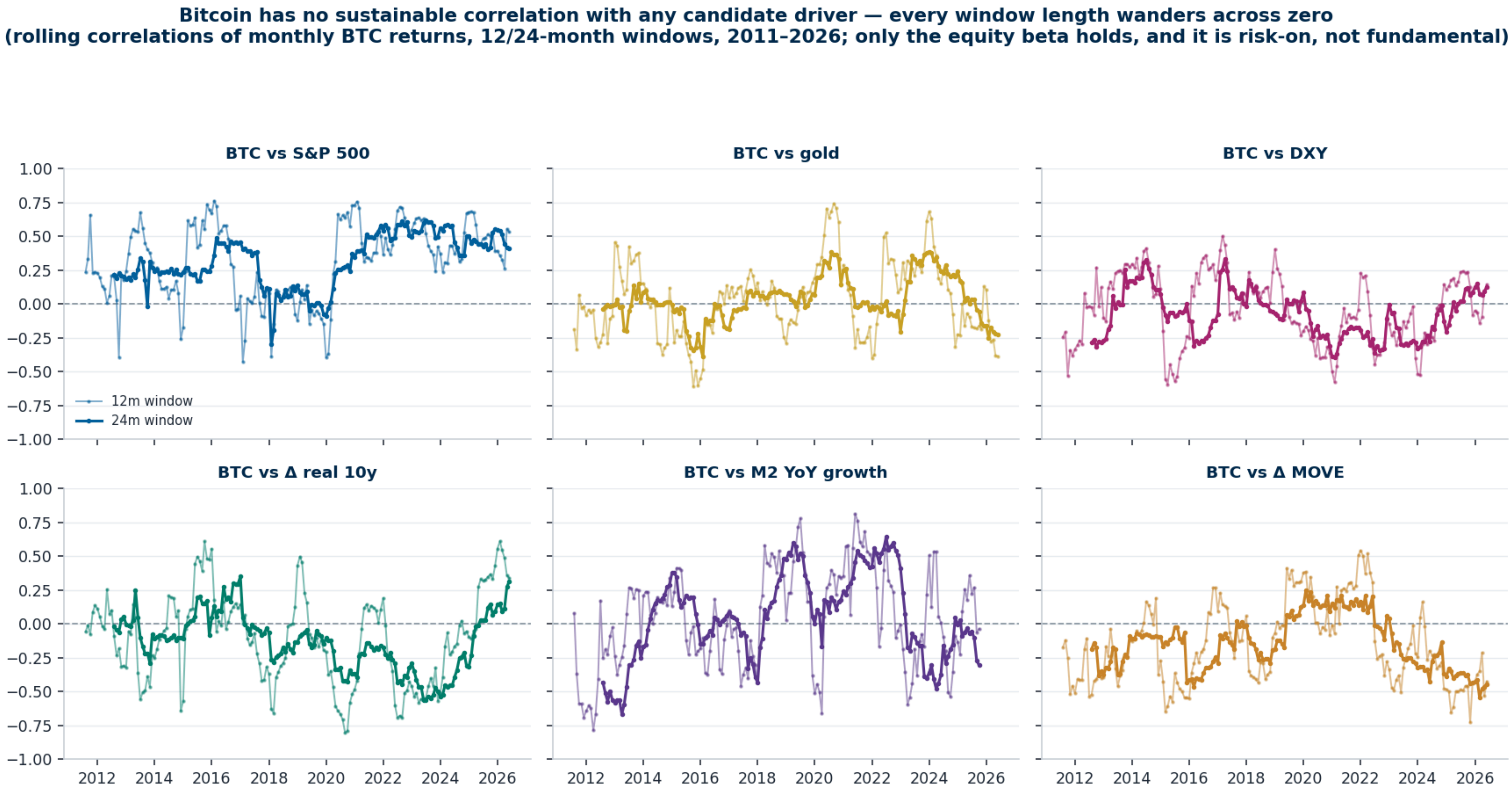
Rolling 5-year nominal total returns by asset, 1976–2025 — S&P worst-quartile windows bolded across all rows

Source(s): JAOD analysis; holders/returns_dist (Shiller, MeasuringWorth, FRED, Ken French). Interactive dashboard (this view): jhdavis789.github.io/gold-holders-dashboard/.

7 I tested twelve candidate drivers: every rolling correlation wanders across zero at both 12- and 24-month windows

- Six macro candidates below; the extended set (Nasdaq, VIX, Fed balance sheet, world M2, active addresses, hashrate) behaves the same — pages 3–4 of the chartpack. The only survivors, **equity beta and BTC's own network activity**, are risk appetite and reflexivity, not forecastable fundamentals.

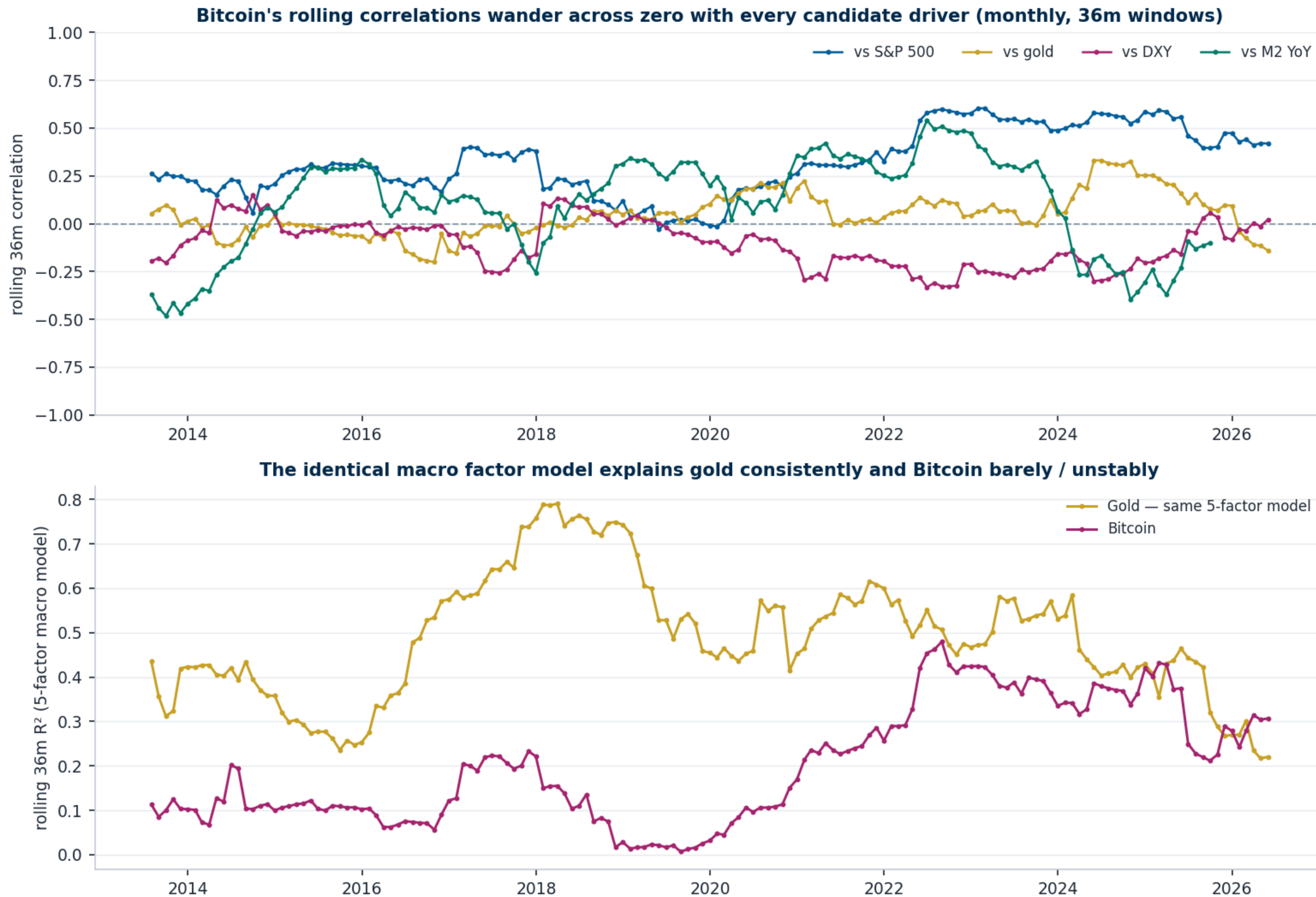
Rolling correlations of monthly BTC returns, 12m (thin) and 24m (thick) windows, 2011–2026



7 The identical five-factor model explains gold consistently and Bitcoin barely — the inconstancy is Bitcoin's, not the method's

- Subsamples 2011-14 / 15-18 / 19-21 / 22-25: gold's real-rate beta keeps its sign with t-stats -3.1 / -5.8 / -6.7 / -2.3; Bitcoin keeps **0 of 5** factors. Same data, same method, same windows.

Top: BTC rolling 36m correlations. Bottom: rolling R^2 of the same 5-factor model — gold (gold line) vs Bitcoin (magenta)



7 My suggested framework owns the inconsistency: a share of gold's monetary pool with the share as the named unknown, times a survival mixture

• With nothing stable, any regression number would be a sample artifact. The industry's own frame (CoinShares, ARK, VanEck) is share-of-a-pool — I make it honest by **naming the share as the assumption** and adding a survival branch (a 20y horizon carries real ruin probability, partly offset by Lindy).

Why this frame

- The gold CMA sizes the pool; the BTC number is a distribution of claims on it — built on the gold work, not beside it.
- The grid spans 3–60% share because that is the honest width of the disagreement; a single point would manufacture precision.
- Feedback I want: does the grid go to the group as a distribution, or as one probability-weighted point with the mixture disclosed? And the survival probability itself.

• At spot $107,754, pool ≈ $23T, supply 19.9M coins:

Scenario grid (before the survival mixture)

Scenario	Share of pool	Implied price	20y CAGR
Bear	3%	$34,673	-5.5%
Base	**15%**	**$173,367**	**+2.4%**
Bull	60%	$693,467	+9.8%

Source(s): BTC one-pager (attached); scenario engine in the evidence deck.

Where I want your feedback before this goes to the group — six treatments, my suggestion on each, plus the two one-pagers

Six areas, my suggestion on each

#	My suggested treatment	The question for you
1	Drift = growth band **4.8–6.7%/yr** by anchor (identity: asset growth − supply growth)	which anchor to carry; the 2pp spread is the bull/bear gap
2	**Valuation overlay** on the ~20y share cycle; lower half of band at the 97th pctile	functional form (percentile haircut vs half-reversion vs scenario band)
3	Dollar leg = **one floating DM index**; Asia buyer/holder basket monitored yearly	is a regime-dependent leg worth the complexity if 2022+ holds?
4	Rate leg = **US 10y real** for now	too dollar-centric? run the global-real-rate study next?
5	CPI regime threshold at **4%** (t +1.9/+2.4 qtrly by spec)	vs 2.5%; it is a live input cell in the workbook
6	BTC = **scenario grid × survival mixture**, share named as the assumption	distribution vs one weighted point; the survival probability

- Everything is one link deep; the attachments are the documents of record.

Links & attachments

Evidence deck (10 slides, full flow)

Papers & shops → factors → approaches (interactive)

Factor model dashboard (in-sample + OOS)

Intercept anchor study

Dollar-leg one-pager

Gold through every drawdown

Rolling-returns tool (bottom-quartile view)

Bitcoin driver-instability chartpack

Portal (everything)

Attached: gold one-pager · BTC one-pager · model workbook · full paper. 30 minutes covers all six.

Source(s): Asks map to the paper's 'Open questions, by research area' section (47pp, attached).

Same signs at every frequency; R^2 rises with horizon

Quarterly

	Dependent variable: gold return (log, %, quarterly)		
	M1	M3	M6 (M3 + mean reversion)
Δ Real 10y (per +1pp)	-9.947*** (-5.91)	-12.737*** (-6.33)	-14.638*** (-6.48)
DXY return (per +1%)	-0.474*** (-2.87)	-0.523*** (-3.21)	-0.495*** (-2.67)
Δ 10Y breakeven (pp)	1.268 (0.68)	2.686 (1.40)	2.352 (1.16)
Δ MOVE (bp)		0.051 (1.54)	0.045 (1.20)
Δ Real10 × High-CPI		6.808* (1.94)	9.156** (2.36)
Gold gap vs 5y avg, lagged			0.011 (0.37)
Constant (α, annualized %/yr)	10.84	10.14	7.39
Observations	93	93	74
Adjusted R^2	0.395	0.421	0.461

*t-statistics (OLS) in parentheses. *p<0.10, **p<0.05, ***p<0.01. HAC inference in the paper.*

Weekly

	Dependent variable: gold return (log, %, weekly)		
	M1	M3	M6 (M3 + mean reversion)
Δ Real 10y (per +1pp)	-2.223*** (-3.71)	-4.111*** (-6.19)	-5.341*** (-7.22)
DXY return (per +1%)	-0.963*** (-15.56)	-0.986*** (-16.17)	-0.963*** (-14.14)
Δ 10Y breakeven (pp)	0.629 (0.77)	1.530* (1.87)	1.227 (1.41)
Δ MOVE (bp)		0.041*** (5.62)	0.049*** (6.20)
Δ Real10 × High-CPI		5.138*** (3.81)	6.177*** (4.41)
Gold gap vs 5y avg, lagged			0.000 (0.01)
Constant (α, annualized %/yr)	10.26	9.66	8.86
Observations	1,221	1,221	962
Adjusted R^2	0.212	0.241	0.251

*t-statistics (OLS) in parentheses. *p<0.10, **p<0.05, ***p<0.01. HAC inference in the paper.*

Source(s): JAOD analysis — gold_models_bbg.xlsx; monthly is the workhorse, these are consistency checks.

A cross-desk check: the inflation-swap specification replicates on identical N — the two desks see the same model

Simple returns on real-rate (bp), DXY (%), 2y inflation swap (bp), 2004-07→

Simple returns, bp	Daily	Weekly	Monthly
Intercept (per period)	0.0490	0.2506	1.0715
Real 10y (per +1bp)	-0.0179	-0.0229	-0.0636
DXY (per +1%)	-0.6874	-0.9353	-0.7196
2y inflation swap (per +1bp)	0.0074	0.0161	0.0117
R^2	0.105	0.219	0.277
N	5720	1144	263

• Remaining daily differences are a data-timing artifact: the LBMA PM fix (2:30pm London) vs NY close splits a day's move across two rows; summing the contemporaneous and one-day-lag betas recovers the NY-close values.

Notes

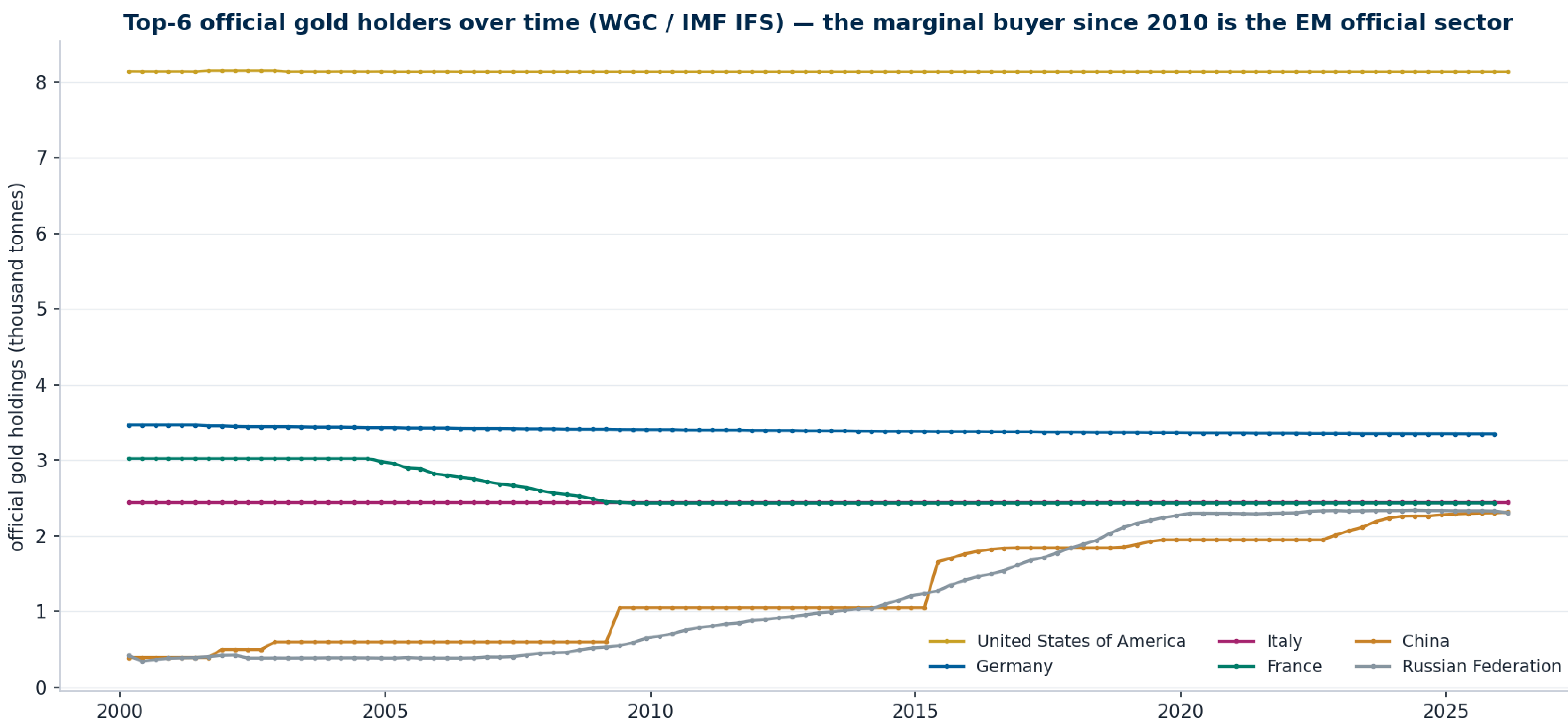
- The swap leg is significant but adds little beyond breakevens at the monthly horizon.
- Full replication tab in the attached workbook.

Source(s): JAOD analysis — workbook replication tab; LBMA PM fix vs NY close timing.

Top-6 official holders over time: the West holds, the East accumulates — the utility-of-owners story behind the drift and the dollar-leg signal

- US / Germany / Italy / France static for two decades; China and Russia are the accumulation. The price-insensitive buyer since 2010 is the EM official sector — the owner-utility lens in one chart.

Official gold holdings, thousand tonnes, quarterly since 2000



Source(s): JAOD analysis of WGC / IMF IFS data. Interactive holders dashboard: jhdavis789.github.io/gold-holders-dashboard/.

The top-down frame behind the drift question: gold's price is a chain of shares of world wealth, divided by a slowly growing stock of ounces

- Written in logs the funnel is an additive expected-return identity — which is what a CMA needs. The rationale for the total-asset anchor: the demand for a store of value scales with the pool of wealth that needs insuring, so through cycles the hedge bucket grows with total assets; if gold failed to keep pace, the market would be pricing a shrinking need for the hedge. Two caveats keep it honest: central banks scale off FX reserves (not global wealth), and a share is an outcome, not a driver — the engine is the marginal flow.

The identity: price = wealth × hedge share × gold-of-hedges share ÷ ounces

$$P = (W \times s_1 \times s_2) \div Q$$

$$\partial \ln P = \textbf{nominal wealth growth} + \Delta\, \textbf{hedge share} + \Delta\, \textbf{gold-of-hedges share} - \textbf{supply growth}$$

Two reference cases fall out cleanly. Share-constant: hold both share terms flat — gold keeps its current slice — and E[return] = (population + productivity + inflation) – ~1.6% supply ≈ **~3–4% nominal / +0.5–1% real**. But "flat" is not neutral — it holds gold at its **97th-percentile share** (see anchors below). The honest mean-reversion neutral lets $\Delta\ln(s_1 \cdot s_2)$ run *negative* toward the historical median, pulling the base case **below** that. Your active thesis (debasement, distrust, real rates, war) is the gap between the two — a scenario overlay, not a free-handed target.

- Three buyers, three objective functions — each keys off a different layer of the funnel. This is the owner-utility lens: the official sector answers to sanctions and reserve concentration, EM households to home-currency debasement, allocators to real rates and the dollar.

Who buys gold, and why

Who buys gold, and why

Three buyers, three objective functions — each keys off a different part of the funnel above.

	CENTRAL BANKS **Official reserve managers**	INDIVIDUALS **Households, especially EM**	INVESTORS **Funds & allocators**
Goal	A reserve no one can freeze, default on, or debase.	Savings that outlast their own government and currency.	Make money, and own the hedge stocks and bonds can't provide.
Core drivers what matters	Sanction & asset-freeze risk USD-reserve concentration Geopolitical fragmentation Peer central-bank flows Scales off the size of FX reserves — not global wealth.	Home-currency debasement & inflation Trust in own government & courts Capital-control risk Cultural & precautionary demand	10y real rate DFII10 · β −0.069 USD direction real corr −0.72 Breakeven inflation β +0.021 ETF flows & momentum BTC as substitute

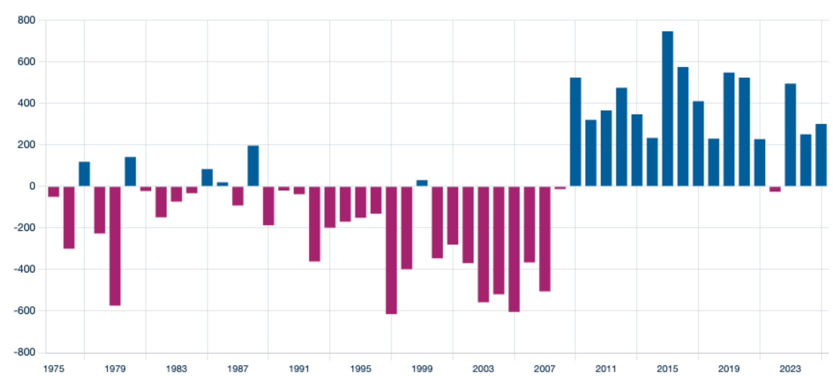
Source(s): JAOD analysis — cma_framework (full page with references in the repo); WGC/IMF holder data.

The official sector flipped from decades of net selling to sustained buying after 2008 — and the West's grip on official gold has loosened ~25 points in 50 years

Year-over-year change in world central-bank gold (net tonnes)

buckets show who is driving the shift; the individual-country panels at the end give the detail behind the buckets.

Chart 5. Year-over-year change in world central-bank gold (net tonnes)



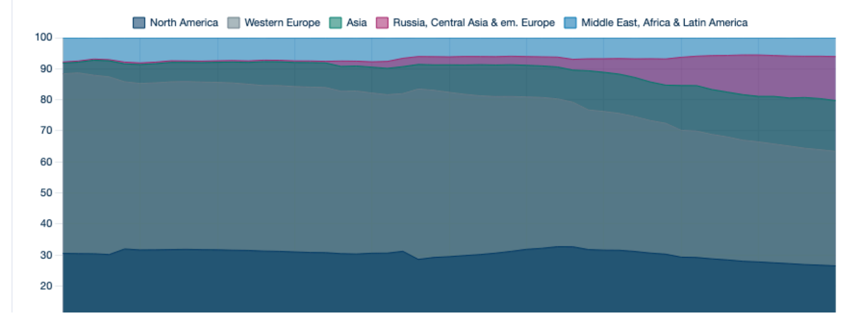
Net change in world official-sector gold each year (last 50). Decades of net central-bank SELLING (European disposals under the 1999–2009 Central Bank Gold Agreements, the troughs below zero) flipped to sustained BUYING after the 2008 crisis and

Share of official gold by major geography (%)

Official gold by major geography, last 50 years. The Western blocs barely move — North America (9,341→8,254 t, essentially the static U.S. hoard) and Western Europe (17,742→11,499 t, dipping during the 1999–2009 Central Bank Gold Agreement sales). All the growth is in the diversifiers: Asia rose 1,052→5,103 t and Russia/Central Asia/emerging Europe 111→4,418 t — the buckets doing the buying.

Source: IMF International Liquidity (IFS), annual gold holdings summed within geographic buckets (reporting central banks; the IMF/ECB/BIS aggregate is excluded).

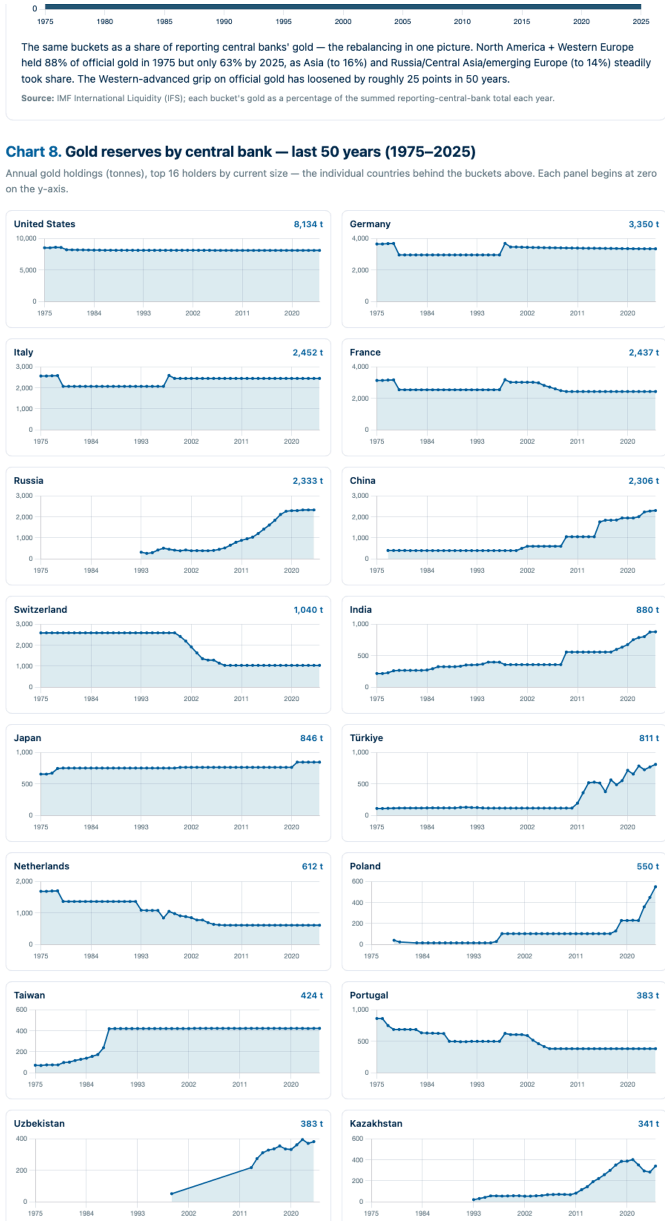
Chart 7. Share of official gold by major geography (%)



Source(s): JAOD analysis — IMF International Liquidity (IFS) reserves data; from the USD-reserve-currency study (10pp PDF in repo).

Gold reserves by central bank, last 50 years: the West static, Russia/China/India accumulating

Top holders, annual tonnes, 1975–2025 — each panel starts at zero



The same buckets as a share of reporting central banks' gold — the rebalancing in one picture. North America + Western Europe held 88% of official gold in 1975 but only 63% by 2025, as Asia (to 16%) and Russia/Central Asia/emerging Europe (to 14%) steadily took share. The Western-advanced grip on official gold has loosened by roughly 25 points in 50 years.

Source: IMF International Liquidity (IFS); each bucket's gold as a percentage of the summed reporting-central-bank total each year.

Chart 8. **Gold reserves by central bank — last 50 years (1975–2025)**

Annual gold holdings (tonnes), top 16 holders by current size — the individual countries behind the buckets above. Each panel begins at zero on the y-axis.

Sources for the gold panels (Charts 5–8). World Gold Council — Gold Reserves by Country (gold.org/goldhub), underlying data IMF International Financial Statistics (IFS) / International Liquidity. Annual world official-sector volume and the per-bank series are from the IMF IFS reserves chapter (data.imf.org — IMF.STA:IL). Geographic buckets sum the reporting central banks within each region (the IMF/ECB/BIS aggregate is excluded); shares are of that summed reporting-country total. Year-over-year change is the first difference of the annual world total, which uses last-reported carry-forward per country so lagging reporters do not depress the most recent observation.

Source(s): JAOD analysis — WGC / IMF IFS per-country holdings; full 16-country grid in the reserve-currency study.

The same five-factor model: median rolling R^2 is 0.47 for gold vs 0.19 for Bitcoin — and every BTC correlation crosses zero repeatedly

- Rolling 36-month correlations of monthly BTC returns: the mean is near zero for every candidate, the range spans both signs, and the sign flips count how often each relationship reversed — a real driver does not cross zero every few years.

Correlation consistency, 2011–2026

Pair (36m rolling)	Mean corr	Range	Sign flips
BTC vs S&P 500	+0.32	-0.03 .. +0.60	4
BTC vs Gold	+0.05	-0.20 .. +0.33	17
BTC vs DXY	-0.11	-0.33 .. +0.15	13
BTC vs Δ real 10y	-0.15	-0.48 .. +0.17	7
BTC vs CPI YoY	-0.17	-0.52 .. +0.41	8
BTC vs M2 YoY	+0.10	-0.48 .. +0.54	4

Identical 5-factor model (Δreal10, DXY, ΔBE10, ΔMOVE, S&P): median rolling R^2 0.47 for gold vs 0.19 for BTC; gold keeps 2 of 5 factors stable across subsamples, BTC keeps 0.

Same regression, four subsamples — β (t) by period

Same five-factor regression, four subsamples: gold has a spine, Bitcoin does not (HAC t-stats)

BITCOIN — β (t-stat) by subsample: signs flip, nothing stays significant

Subsample	Δ Real 10y	DXY	Δ BE10	Δ MOVE	S&P 500	R^2	N
2011-14	+8.07 (+0.2)	-0.10 (-0.0)	-18.65 (-0.3)	-8.35 (-1.4)	+2.41 (+0.9)	0.07	48
2015-18	+16.66 (+0.8)	-0.17 (-0.1)	-8.57 (-0.3)	-3.87 (-1.0)	+1.34 (+1.5)	0.09	48
2019-21	-2.47 (-0.2)	-1.01 (-0.5)	+3.65 (+0.2)	+5.46 (+3.1)	+1.75 (+2.9)	0.26	36
2022-25	+6.02 (+0.5)	-0.31 (-0.2)	+32.45 (+1.8)	-1.61 (-1.4)	+1.57 (+2.6)	0.38	48

GOLD, identical regression — the real-rate and dollar betas keep their sign in every subsample

Subsample	Δ Real 10y	DXY	Δ BE10	Δ MOVE	S&P 500	R^2	N
2011-14	-15.91 (-3.1)	-1.11 (-1.9)	-1.25 (-0.1)	+0.88 (+1.0)	+0.16 (+0.8)	0.36	48
2015-18	-15.13 (-5.8)	-0.94 (-6.1)	-5.89 (-1.5)	+2.07 (+4.6)	+0.05 (+0.3)	0.67	48
2019-21	-16.04 (-6.7)	-1.19 (-3.5)	+3.71 (+0.6)	-0.22 (-1.0)	-0.29 (-1.2)	0.60	36
2022-25	-4.83 (-2.3)	-0.85 (-3.2)	+1.75 (+0.4)	+0.34 (+0.9)	-0.25 (-2.0)	0.34	48

Source(s): JAOD analysis — BTC driver-instability study (results.json; chartpack p6).